

DSC5001: Statistical Machine Learning I

Assignment 1

Semester A, 2026/27

Name: _____

Student ID: _____

Total: 100 marks

- There are eight questions.
- Questions 1–4 are calculation problems worth 50 marks in total.
- Questions 5–8 are derivation problems worth 50 marks in total.
- Questions 1 and 5 correspond to Week 1; Questions 2 and 6 to Week 2; Questions 3 and 7 to Week 3; and Questions 4 and 8 to Week 4.
- Show the essential reasoning leading to each answer.
- Numerical answers without supporting work may receive limited credit.
- Unless a formula is explicitly supplied, use the definitions and results developed in the lectures and tutorials.

Question	1	2	3	4	5	6	7	8	Total
Marks	10	10	10	20	10	10	10	20	100

Question 1: Gaussian Geometry and Linear Combinations [10 marks]

Let

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 & 3 \\ 3 & 4 \end{bmatrix} \right).$$

Consider the two observations

$$\mathbf{x}_A = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \quad \mathbf{x}_B = \begin{bmatrix} 2 \\ -2 \end{bmatrix}.$$

- (a) Compute the squared Mahalanobis distance of each observation from the Gaussian mean. [3 marks]
- (b) Determine which observation has the larger Gaussian density and compute

$$\frac{p(\mathbf{x}_A)}{p(\mathbf{x}_B)}.$$

Your argument should use the multivariate Gaussian model rather than Euclidean distance alone. [3 marks]

- (c) Define

$$U = X_1 + X_2, \quad V = X_1 - X_2.$$

Compute $\text{Var}(U)$, $\text{Var}(V)$, and $\text{Cov}(U, V)$. Are U and V independent? Justify your answer. [4 marks]

Question 2: Method of Moments versus Maximum Likelihood [10 marks]

Suppose $X_1, \dots, X_n$ are independent observations from the model

$$f(x \mid \theta) = \begin{cases} \frac{2x}{\theta^2}, & 0 \leq x \leq \theta, \\ 0, & \text{otherwise,} \end{cases} \quad \theta > 0.$$

The observed sample is

1, 2, 3, 4.

- (a) Using the first population moment, obtain the Method-of-Moments estimate of θ . [3 marks]
- (b) Construct the likelihood as a function of θ and obtain the maximum-likelihood estimate. Pay particular attention to the support of the model. [4 marks]
- (c) Compare the two estimates. Explain why one of them assigns zero likelihood to the observed dataset, and state what this example shows about mechanically matching moments. [3 marks]

Question 3: Bootstrap Uncertainty for a Range Statistic [10 marks]

The observed sample is

$$D = \{1, 3, 4, 8, 10\},$$

and the statistic of interest is the sample range,

$$T(D) = \max(D) - \min(D).$$

Eight bootstrap resamples are shown below:

1	(1, 3, 3, 4, 8)	5	(1, 1, 3, 4, 4)
2	(1, 1, 4, 10, 10)	6	(3, 4, 8, 8, 10)
3	(3, 3, 4, 8, 8)	7	(1, 3, 8, 10, 10)
4	(1, 4, 4, 8, 10)	8	(1, 3, 4, 4, 8)

- Compute the observed statistic and the eight bootstrap replicate values. [3 marks]
- From these replicates, compute the bootstrap bias estimate and bootstrap standard error of the range. [4 marks]
- A second analyst proposes drawing five observations *without replacement* from D in every repetition. Explain what would happen to the estimated uncertainty of the range and why this is not the ordinary nonparametric bootstrap used in the course. [3 marks]

Question 4: Regression Fitting, Leverage, and Influence [20 marks]

Consider the simple linear regression dataset

$$x = (0, 1, 2, 3, 4, 9), \quad y = (1, 3, 8, 7, 9, 18).$$

Fit a linear model with an intercept.

- (a) Obtain the OLS fitted line. Show the quantities needed to obtain the intercept and slope rather than quoting a software output. [5 marks]
- (b) Compute the fitted values, residuals, residual sum of squares, and the usual estimate of the error variance. [4 marks]
- (c) Compute the leverage of every observation and identify the highest-leverage point. [4 marks]
- (d) Using the Cook's-distance expression developed in Week 4, compute an influence score for every observation, up to the common positive scaling factor used in the lecture. Identify the most influential observation. Is it also the observation with the largest absolute residual? [4 marks]
- (e) Remove the most influential observation, refit the regression line, and compare the new slope with the original slope. Explain what this calculation shows about influence. [3 marks]

Question 5: Mean and Covariance under a Linear Transformation
[10 marks]

Let $\mathbf{X} \in \mathbb{R}^d$ be a random vector with finite second moments, mean $\boldsymbol{\mu}$, and covariance matrix $\boldsymbol{\Sigma}$. Let

$$\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b},$$

where $\mathbf{A} \in \mathbb{R}^{m \times d}$ and $\mathbf{b} \in \mathbb{R}^m$ are fixed.

- (a) Starting from the definition of expectation, derive the mean vector of $\mathbf{Y}$. [3 marks]
- (b) Starting from the definition of a covariance matrix, derive the covariance matrix of $\mathbf{Y}$. Do not quote the transformation rule without proof. [5 marks]
- (c) Specialize your result to the scalar random variable

$$Z = \mathbf{a}^\top \mathbf{X}.$$

Explain from your expression why $\text{Var}(Z)$ cannot be negative. [2 marks]

Question 6: Bias–Variance Trade-off between Two Bernoulli Estimators
[10 marks]

Let

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p), \quad S = \sum_{i=1}^n X_i.$$

Consider the two estimators

$$\hat{p}_1 = \frac{S}{n}, \quad \hat{p}_2 = \frac{S+1}{n+2}.$$

- (a) Derive the bias and variance of both estimators. [4 marks]
- (b) Derive the mean-squared error of both estimators. [3 marks]
- (c) Determine, as an inequality involving p and n , when $\hat{p}_2$ has smaller MSE than $\hat{p}_1$. Briefly explain why an estimator with nonzero bias can nevertheless have smaller MSE. [3 marks]

Question 7: Sampling Distribution Induced by the Bootstrap [10 marks]

Let the observed dataset be

$$D = \{x_1, \dots, x_n\}, \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

In the ordinary nonparametric bootstrap, each bootstrap observation is drawn independently from the empirical distribution that places equal probability on the observed values. Let

$$\bar{X}^* = \frac{1}{n} \sum_{j=1}^n X_j^*$$

denote the mean of one bootstrap resample.

- (a) Conditioning on the observed dataset D , derive the mean and variance of one bootstrap draw X_j^* . [4 marks]
- (b) Use your result to derive the conditional mean and variance of $\bar{X}^*$. Hence obtain the bootstrap bias estimate and bootstrap standard error for the sample mean. [4 marks]
- (c) Rewrite the bootstrap variance of $\bar{X}^*$ in terms of the usual unbiased sample variance. Identify the finite-sample multiplicative factor that appears. [2 marks]

Question 8: Projection Geometry and OLS Variability [20 marks]

Consider the linear regression model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where $\mathbf{X} \in \mathbb{R}^{n \times p}$ has full column rank and

$$\mathbb{E}[\boldsymbol{\varepsilon} \mid \mathbf{X}] = \mathbf{0}, \quad \text{Var}(\boldsymbol{\varepsilon} \mid \mathbf{X}) = \sigma^2 \mathbf{I}.$$

Let $\hat{\mathbf{y}}$ be the OLS fitted vector, $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$ the residual vector, and $\mathbf{H}$ the OLS hat matrix.

- (a) Write $\mathbf{H}$ in terms of $\mathbf{X}$ and prove that it is both symmetric and idempotent. [5 marks]
- (b) Prove that the residual vector is orthogonal to every column of $\mathbf{X}$. Then derive

$$\sum_{i=1}^n h_{ii} = p,$$

where h_{ii} denotes the i th leverage value. [5 marks]

- (c) Derive the conditional covariance matrices of the fitted vector $\hat{\mathbf{y}}$ and the residual vector $\mathbf{e}$. [6 marks]
- (d) Use the diagonal entries of your results from part (c) to describe how leverage affects the variability of a fitted value and the variability of its residual. Explain why this matters when interpreting unusually large residuals. [4 marks]